MS. NO. 181412 ENTITLED "NETWORKS OF VIOLENCE: PREDICTING CONFLICT IN NIGERIA"

Dear Professor Gailmard,

Thank you for the opportunity to revise and resubmit our manuscript. We believe the manuscript has benefited from the Reviewers' helpful and thoughtful comments. We have revised the manuscript, taking seriously each individual point raised by the Reviewers. The revision memo is organized by first responding to your comments and then addressing the reviewers' points. Our comments and responses are shown in *BLUE* below each point.

We hope you agree that the manuscript has improved through this process and we are looking forward to your response.

Sincerely,

The Authors.

Date: July 5, 2018 Version 0.01.

1. Editor

(1) Though this piece came to me as a methods paper, I agree with R. 2 that the best approach is probably to focus foremost on the substantive motivation, lay out your original theoretical perspective, show how that determines the choice of methodology, and explain what you found. That dovetails with R. 3's good suggestions about approaching the presentation, and with R. 1's critiques about the clarity of the central aim of the paper. The key questions are boiled down succinctly by R. 3 who asks that you clarify, "what are the consequences of these limitations [that you identify in previous approaches]? What don't we understand that, at the end of this article, we will understand better?"

- *We clarify the purpose of the paper, principally through a complete re-write of the introduction and theoretical section. In the introduction we now begin by explaining how existing dyadic analyses of conflict, which tend to focus primarily on non-state actor vs. state dyads, miss important actors and interactions relevant to understanding conflict processes as whole. We illustrate this point with historical anecdotes. In the new theoretical section we now begin with first principles on the motivations and attributes of armed groups, drawn from the literature, to explain violent behavior during conflicts. While the focus on the rebel-state dyad is important, it overlooks how interactions between actors might inform future behavior. We then explain how a network approach flows from the motivations and attributes in the dyadic literature, and how this approach allows us to systematically examine the influence of relationships and actors' interactions on levels of conflict over time.*

(2) In addition, though you certainly sought to avoid wading into the latent space "vs." (T)ERGM "debate", I agree with R. 1 that more discussion of your preference for one over the other is called for. To me the most straightforward (if labor intensive), and most transparent way to handle this is to consider TERGM as a robustness check in an appendix.

- *Insert great response.*

2. Reviewer 1

2.1. **Major Comments.**

(1) The research objective of this paper should be clarified. I can see four different aims that are addressed at least partially in this paper. The first is to develop the best performing classifier to predict battles in a civil conflict. The second is to understand whether a latent variable network model performs better than GLM in predicting battles, with the intent of interpreting the performance gain as the benefit of accounting for network dependencies. The third is to develop an explanatory model of battles from which we can learn about the factors that serve as indicators of civil conflict. The first is to propose and illustrate the application of a latent variable network model that can incorporate exogenous variation in the vertex set. Once the authors clarify their objective, it would be useful to provide additional analysis that builds toward their aim. For example, if the goal is to develop an effective classifier, it would be valuable to see comparisons to random forests, neural networks, and other learning approaches at are designed to optimize predictive performance.

- *Thank you for this useful feedback. We hope that our revisions throughout the paper clarify the contributions of the paper. Specifically, in-line with the editor's comments, we have re-written the front end of the paper to better articulate how interactions between actors help to explain the occurrence of conflict over time and contribute to the broader literature on conflict processes. With this said, we continue to also emphasize the advantages of our approach, in terms of predictive performance and classification. As such we have added a section to the appendix in which we compare the performance of AME with a random forest model. To evaluate performance, we utilize the same cross-validation procedure as was already presented in the paper. The results are shown in Section A.1.6 of the Appendix. We have also included a visualization of the results below for convenience.*

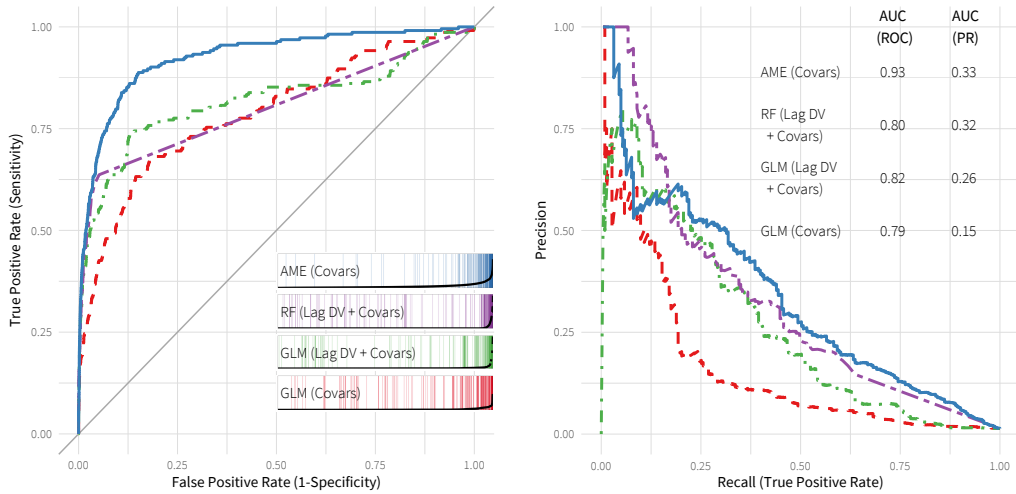

FIGURE 1. Assessments of out-of-sample predictive performance using ROC curves, separation plots, and PR curves. AUC statistics are provided as well for both curves. In each curve, the AME model is in blue, RF is denoted in purple, the GLM with lagged DV and covariates is in green, the GLM with only covariates is in red.

(2) Second, the empirical modeling presented in this paper is generally very carefully executed. One aspect of the model specification that I would recommend exploring is the degree to which there is temporal heterogeneity in the parameter values. I am not an expert in civil conflict, but it seems likely that the factors governing battle instances will shift over time throughout the conflict.
   - *Insert great response.*

(3) I recommend that the authors add the TERGM as a comparison model, or provide additional justification for not comparing their model to TERGM. For example, perhaps after the goals of the paper are clarified, it would be clear that there would be no need to compare their model to yet another network model that can account for dependence ... it would be great to see a robustness check in which the authors evaluate whether the effects of covariates are robust to modeling with TERGM.
   - *Insert great response.*

2.2. **Minor Comments.**

(1) I would like to see a discussion of setting the priors used for inference with this model. In particular, how does the specification of the priors relate to the predictive performance of the latent variable model? If the priors make any sort of difference, how should the priors be set to assure that the model performs as well as possible while also assuring that the model is not trained on the test data.
   - *We discuss the priors on page ...*

(2) Second, for the predictive experiment, it would be great to see if the comparison of the AME and GLM is sensitive to k—the number of groups into which the data is split ... it would be great to see if the predictive performance of AME relative to GLM varied between, say k=5 and k=20.

- *We insert a mention of the value of k used in our analysis on page X. We also, in the appendix, compare the predictive value for values of k ranging from 5 to 20.*

3. Reviewer 2

3.1. **Major Comments.**

(1) What I find interesting and publication-worthy about this paper is the substantive insights about actors, their effects of one another, and the entry of new conflict participants. This is the part that constitutes the core of the contribution, not the methodological approach. I would support publication of this manuscript as a substantive piece that is driven by an empirical puzzle and analyzed using an appropriate method. The manuscript, as it is currently written, sells itself as methodological fix rather than a substantive advance on our knowledge of civil war dynamics. I think this is a poor framing/presentational choice and sells the contribution short. It also seems backwards. I would start with a substantive puzzle, build a theoretical model of interaction among actors of different types, derive predictions, and only then get to the method and testing. As it is, the actual substantive story does not start until p. 8–this is where I would start the paper.

- *Thank you for this helpful feedback! We appreciate your interpretation of our contribution and have worked hard to use these core insights to rewrite the front end of the paper. Specifically, we have rewritten the introduction to better set up the primary empirical puzzle of "Who fights who and when?" In the new theoretical section we now begin with first principles on the motivations and attributes of armed groups, drawn from the literature, to explain violent behavior during conflicts. We then explain how a network approach flows from the motivations and attributes in the dyadic literature, and how this approach allows us to systematically examine the influence of relationships and actors' interactions on levels of conflict over time. In doing so, we have also more deeply incorporated literature by Will Moore and others who have highlighted the importance of interactions to explain violent behavior.*

(2) I especially like the part about new participant entry. Why would we expect that an entry by a new participant would affect the conflict dynamics? What types of participants are likely to enter an ongoing conflict? What possible relationships may a new participant have to the current participants? Why would an entry of a new participant increase outbidding? I think the authors have some interesting ideas, but these need to be developed much further. I would like to be presented with a well-grounded theoretical story related to the entry of a new participant in an ongoing conflict.

- *We attempt to address all of these questions about new actor entry.*
  - *As we discuss on page 7, entries by a new participant can increase the likelihood of conflict by weakening the government, leading to the government becoming more violent to restore credibility, or by imperiling the resource bases of rebel actors.*
  - *As we address in footnote 4, page 8, we do believe that entrants in a civil conflict will not, on average, be more powerful or violent than groups already in a conflict, but that the entrants of particularly strong and violent actors are what risk an escalation.*
  - *We discuss on pages 8-9 how a new entrant into a civil conflict could be tacit allies with certain groups, attack other groups, and have no direct relation to some groups.*

  - *We discuss (first paragraph, page 10) how outbidding in particular is likely to happen to groups that both rely on civilians for support and have similar constituencies, when a new group is stronger, more aggressive, or more successful than existing groups.*

(3) Instead, the current draft spends the first 7 pages trying to convince the reader that the study of civil war is important and that actors in a civil war may affect one another's actions–both of these ideas are non-controversial. Then there is a lengthy discussion of network terms and methods, which I would move to the research design section. I would also move the history of Nigerian conflict to the appendix, is distracts from the flow.

- *We addressed this feedback in three ways. First, we shortened the introduction and re-wrote it it to be concise, clearer about the goals of the paper and, we hope, more compelling. Second, we removed the section on network terms (including the "stylized network" graphic) from the main body of the paper and added it to the appendix. Given the substantive orientation of the revised paper, we now see this section as distracting, though useful for those unfamiliar with network terminology. Third, we have moved, edited and shortened the section summarizing the Nigerian case. We now present this information as part of our Data section, because we believe it is important to discuss the case in order to illuminate both our case selection justification as well as provide context for our model and, later, our findings. We hope that these changes remedy the paper's flow problem.*

4. Reviewer 3

4.1. **Major Comments.**

(1) It is not entirely clear at any point in this manuscript what problem the manuscript is trying to address. The author(s) are correct that civil war scholars have not properly considered the interdependencies among actors in multiparty conflicts and that dyadic research designs are limited. However, what are the consequences of these limitations? What don't we understand that, at the end of this article, we will understand better?

- *By re-writing the introduction, theory, and concluding sections, we hope that the main point of the paper is much clearer. We focus on the question of "Who fights who and when" in the context of multi-group armed conflicts. We argue, and find support, for the position that the interactions, or behavior, of armed actors are just as important for explaining future occurrences of violence as armed group attributes. Throughout the paper, but especially in the introduction, we suggest that by missing the interactions present in violent conflict scholars will mischaracterize patterns of violence and ignore extra-dyadic factors that make violence between groups more likely.*

(2) The author(s) state on page 3 "This study investigates how relationships between armed actors evolve over time, and how the structure of relationships in one time period influences the occurrence of violence between armed actors in a future period. To do this, we consider how interdependence across actors, changes in actor composition over time, and actor-level attributes predict violence in the network." There are several potential questions here, such as whether different armed groups fight (or cooperate), how the presence of additional actors affects the level of violence between rebel groups and the state, the overall level of violence in the conflict, etc. The lack of specificity comes through in the theory, where there is a discussion of key concepts in network theory like actor centrality and reciprocity, but it is unclear how exactly these concepts relate to the question in the manuscript or to existing work. It is also present in the analysis, where the DV is the occurrence of battles between armed actors, but it is not clear why that is appropriate given the theory.

- *We have now re-written the introduction and the theory sections. First, we hope that our introduction is much clearer and orients the paper around the question "who fights who and when"? We illuminate this puzzle through providing anecdotes about different multi-actor conflicts in our new introduction. Our theory section is now rooted in contributions from prior research on the attributes and motivations of armed actors. Such a focus allows us to then explain how a network approach flows from the findings of preexisting research. Each network concept used in the theory is now situated within the related literature on conflict processes, which allows us to defend our position that the interactions between actors, not just actor attributes, more accurately explains the occurrence of violence between armed groups.*

(3) All of this is fixable, but it requires re-framing the introduction, better situating the argument in the literature, and re-writing the theory. Why not start with the question why do some armed groups fight each other and others do not? There is clear literature on this question, much (although not all) of which is cited here. Describe that literature and explain why it is insufficient to understand this phenomenon.

Then, the manuscript needs a clear, focused theoretical argument drawing on network theory to explain this. The theory should start from first principles what are armed actors trying to achieve through violence and how do the components here (centrality, reciprocity, the introduction of new "aggressive" actors, etc.) fit into this story.

- *We agree with this feedback and, as noted above, have completely rewritten the front end of the paper in an effort to clarify our focus on this question of why some groups fight each other. We have also retitled the paper to reflect this focus.*

4.2. **Minor Comments.**

(1) To identify groups, the author(s) use ACLED. I understand why, but the downside of doing this is that they lose the ability to include group level characteristics present in, for example, the Non-State Actor data, the EPR project, etc. The authors need to make a clearer case for the contribution here given that they are giving up substantial access to information about these actors.

- *We make the case for our use of the ACLED data on page 14, footnote 11. Our general motivation for using the ACLED data is that more actors are included in the data because there is no threshold requirement, such as a specific number of battle-related deaths, like there is in the UCDP data. This is useful for a study interested in explaining fighting between actors since not all battles will result in a high number of deaths, but all battles might influence the trajectory of the conflict.*

(2) The manuscript needs to be much clearer on what the system is. Here, it includes everyone fighting in Nigeria. But is that always appropriate? Is the country the box that bounds networks? Why? What about separate, peripheral, conflicts in places like India, should they be modelled together? What about linked conflicts in places like Central Africa? Should they be modelled separately? These considerations are important.

- *We discuss how the modeling framework would differ given transnational or subnational conflicts on page XXX.*

(3) The manuscript should, at the least, cite Bapat and Bond (2011), which is one of the earliest articles to examine cooperation/conflict between rebel groups in civil war.

- *We thank you for pointing out this oversight. We now discuss Bapat and Bond's paper on page 7.*